

Smooth muscle cells of intracranial vessels: from development to disease

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Abstract

Cerebrovascular diseases that cause ischaemic or haemorrhagic stroke with subsequent loss of life or functional capacity due to damage of the brain tissue are among the leading causes of human suffering and economic burden inflicted by diseases in the developed world. Diseases affecting intracranial vessels are significant contributors to ischaemic and haemorrhagic strokes. Brain arteriovenous malformations, which are a collection of abnormal blood vessels connecting arteries to veins, are the most common cause of intracranial haemorrhage in children and young adults. Saccular intracranial aneurysms, which are pathological saccular dilations mainly occurring at bifurcations of the large intracranial arteries near the circle of Willis, are highly prevalent in the middle-aged population, causing significant anxiety and concern; their rupture, although rare, is a significant cause of intracranial haemorrhage in those past middle age that is associated with a very sinister prognosis. Cerebral small-vessel disease, which comprise all pathological processes affecting vessels <500 microns in diameter, account for the majority of intracerebral haemorrhages and ~25% of ischaemic strokes and 45% of dementias in the elderly. In this review, we summarize the developmental, structural, and functional features of intracranial vessels. We then describe the role of smooth muscle cells in brain arteriovenous malformations, intracranial aneurysms, and small-vessel diseases, and discuss how the peculiar ontogeny, structure, and function of intracranial vessels are related to the development of these diseases.

Keywords

Smooth muscle cells • Intracranial arteriovenous malformations • Intracranial aneurysms • Cerebral small vessel diseases

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1. Introduction

Cerebrovascular diseases that cause loss of life or loss of functional capacity due to death or impaired function of neurons or glial cells are among the leading disease-related causes of human suffering and economic burden in the developed world.¹ Cerebrovascular diseases manifest as various forms of ischaemic or haemorrhagic stroke, resulting from occlusion, dysfunction, or rupture of intracranial arteries, capillaries, or veins. In addition to the type of stroke produced, cerebrovascular diseases can be categorized according to the location (extracranial vs. intracranial) or size (large vs. small) of the affected vessel, and whether they arise from a developmental defect or a degenerative process. The clinical manifestations of cerebrovascular diseases are determined by which neural structures are damaged, and range from paralysis or other focal neurological deficits, to gait impairments and motor control problems, and ultimately cognitive impairment and dementia owing to more

widespread involvement of the brain. A common feature of the clinical presentation of cerebrovascular diseases is the severely limited functional capacity of the patient and a resulting dependence on outside help, often for daily living.

The goal of this review is to describe the role of smooth muscle cells (SMCs) in cerebrovascular diseases that affect intracranial vessels, and to discuss the pathobiological implications arising from the peculiar ontogeny, structure, and function of these vessels, in the context of selected diseases that highly impact different age groups.

2. Cerebrovascular diseases as a public health and clinical problem

Stroke caused by cerebrovascular disease is the second most common cause of death in the developed world, accounting in 2015 for about

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6.3 million deaths—3.0 million from ischaemic stroke and 3.3 million from haemorrhagic stroke.¹

2.1 Ischaemic stroke and vascular dementia

The most common cause of ischaemic stroke is occlusion of a cerebral artery by a clot embolus originating either from the heart or, more commonly, from atherosclerotic lesions or dissections in extracranial arteries.^{2,3} The atherosclerotic lesion or dissection may also obstruct the lumen locally to the extent that it impairs flow or leads to a local thrombosis that occludes the artery.² Another major cause is cerebral small-vessel disease (SVD), which accounts for ~25% of ischaemic strokes and 45% of dementias.^{4,5} The prevalence of vascular dementia is as high as 2.4% in Western countries in individuals over 70 years of age, and significantly increases with age.⁵ SVDs comprise all pathological processes that affect the small leptomeningeal and intraparenchymal arteries and arterioles <500 microns in diameter, as well as capillaries and small veins.⁴ Apart from their very rare acute inflammatory forms, SVDs encompass chronic diseases including cerebral amyloid angiopathy, which is especially prevalent at the latest stages of life, and the non-amyloid SVDs, a larger group of disparate pathologies that are commonly related to vascular risk factors, mainly age and hypertension.⁴

2.2 Haemorrhagic stroke

Although only ~10–20% of strokes are haemorrhagic,^{3,6} the mortality associated with them is comparable to or even exceeds that of ischaemic stroke.¹ The most common form of haemorrhagic stroke in the elderly is spontaneous intracerebral haemorrhage, most often caused by SVDs.^{3,4,6} SVDs are less frequent in the middle-aged population, where subsequent rupture of an intracranial aneurysm (IA) is a relatively more significant cause of haemorrhagic stroke.^{3,6,7} IA is a pathological saccular dilation of a large intracranial artery that usually develops after the age of 40 years.^{8,9} Rupture of these saccular IAs (sIAs), which is associated with a very sinister prognosis (~40% mortality),¹⁰ causes a high-pressure haemorrhage into the subarachnoid space, termed an aneurysmal subarachnoid haemorrhage (aSAH), and sometimes also into the brain parenchyma itself. Although the incidence of intracranial haemorrhage caused by aneurysm rupture is low (~10/100 000 in most populations),⁷ the prevalence of unruptured IAs is many times higher (~3%) in those past middle age.¹¹ Because of their relatively high prevalence, unruptured IAs cause significant anxiety and concern among middle-aged, otherwise healthy populations.

The most common cause of intracranial haemorrhage in children and young adults is an arteriovenous malformation in the brain.¹² Brain arteriovenous malformations (bAVMs), which are most often diagnosed after they have become symptomatic, typically manifest as fulminant haemorrhage or epilepsy due to microhaemorrhages, and are found at an incidence of ~1/100 000.¹³

3. Specific developmental, anatomical, and physiological traits of cerebral vessels

The brain circulation is supplied by four extracranial arteries: two carotid arteries that form the anterior cerebral circulation, and two vertebral arteries that fuse inside the skull and give rise to the posterior cerebral circulation. The anterior and posterior circulations are connected at the base

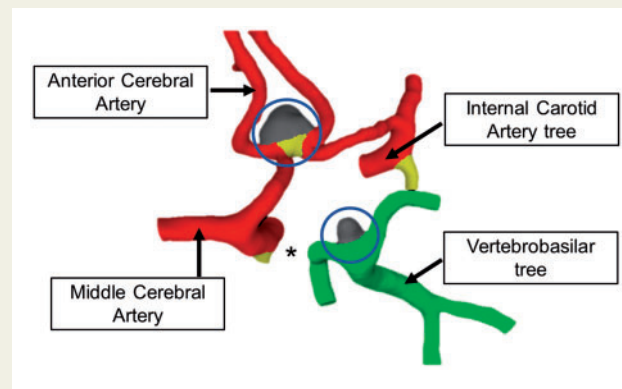


Figure 1 In a reconstruction of the CoW, vessels of the anterior circulation described as being of neural crest origin in chicken are marked in red, whereas vessels of paraxial mesoderm origin are marked in green. Anterior and posterior communicating arteries, marked in yellow, connect the anterior and posterior circulations, and are the most often missing vessels due to agenesis (missing posterior communicating artery marked with *). Most intracranial aneurysms (marked in grey, with a blue circle) arise in the bifurcations of the CoW, although also found relatively commonly in the anterior circulation distally to the CoW.

of the skull by anastomosing arteries that connect the bifurcations from which the major cerebral arteries originate, forming a circulatory continuum of arterial anastomoses called the Circle of Willis (CoW) (Figure 1). The large arteries that originate from the CoW follow the surface of the brain and divide into progressively smaller branches, which then penetrate the brain parenchyma and eventually feed the capillary bed.

3.1 Development of the brain vasculature

An important step in the development of cranial vasculature is the formation of so-called pharyngeal or branchial artery arches, which give rise to the intra- and extracranial vasculature of the head and neck.¹⁴ It is worth noting that the carotid arteries, brachiocephalic artery, and the aortic arch all develop from the third and fourth pharyngeal artery, derived from the neural crest.^{14,15} The internal carotid artery (ICA), which initially supplies both the anterior and posterior circulations, is the first of the large cerebral vessels to develop.¹⁴ Interestingly, in chicken the anterior circulation supplied by the ICA has been shown to arise from the neural crest, whereas the posterior circulation, which develops at a later stage and is supplied by the vertebral arteries, develops from the paraxial mesoderm.¹⁵ In the context of this different ontogeny of the anterior and posterior circulations, it is also interesting to note that in chickens, the neural crest angioblasts that form the primitive vessels originate from the mesoderm.¹⁶

The intracranial vasculature of the human brain starts to develop extracerebrally from a meningeal layer surrounding the early brain.¹⁷ The outer part of this meningeal layer gives rise to dural venous sinuses, whereas the large cerebral arteries that run on the outer surface of the brain develop from the middle layer, and the small arteries that perforate the brain parenchyma as well as the capillary bed develop from the inner layer.¹⁷ The inner or pial layer develops into a pial capillary anastomotic plexus, from which sprouting vascular elements perforate the brain's external glial-limiting membrane, invade into the adjacent neural tissue and further expand via sprouting angiogenesis, giving rise to the small

vessels of the brain parenchyma.¹⁷ Brain angiogenesis is highly dynamic, and the vasculature is further expanded and remodelled throughout both prenatal and postnatal development.^{17,18}

Molecular and cellular mechanisms involved in the development of the brain vasculature have been extensively reviewed,^{18,19} and will be briefly touched on in this review. Whether through vasculogenesis or angiogenesis, vascular morphogenesis—the formation of blood vessels—is composed of the following steps (Figure 2): (1) development of tubes, branching, and anastomosis through endothelial cell proliferation and sprouting; (2) artery or vein specification; and (3) structural maturation, involving recruitment of pericytes and SMCs with subsequent artery wall extracellular matrix (ECM) synthesis.²⁰ While these are all multifactorial processes, vascular endothelial growth factor (VEGF) signalling has a leading role in the first step, Notch signalling in the second, platelet-derived growth factor (PDGF)B and transforming growth factor (TGF)- β signalling in the third step.^{20,21} To induce endothelial proliferation and sprouting, VEGF-A acts through VEGF receptor 2 to induce the activation of a phospholipase-C γ /mitogen-activated protein kinase (MAPK)-mediated signalling pathway that results in the activation of extracellular signal-regulated kinase (ERK) and Notch pathways.²² Notch 1 and the endothelial cell specific Notch 4 signalling upregulate artery specific ephrin B2 in endothelial cells, and through this upregulation and concomitant downregulation of the vein specific EphB4, regulates the differentiation of endothelial cells into an artery or vein.^{20,22} Recruitment, proliferation, and differentiation of mural cells in pericytes or SMCs to newly formed vessels as well as ECM synthesis involve a regulated set of intercellular communication between endothelial and mural cells requiring endothelial-derived PDGFB, which signals through PDGF receptor-beta (PDGFR β) expressed by mural cells, and endothelial-derived TGF β , which signal through a set of receptors expressed by mural cells or endothelial cells, with possible crosstalk between PDGFB and TGF β signalling.^{20,21,23,24} It is worth mentioning that Notch signalling also regulates VEGF-mediated angiogenesis and tube formation²⁵ through regulation of the responsiveness of endothelial cells to pro- and antiangiogenic stimulation by VEGF-A and TGF β , respectively,²⁶ and PDGFB is potentially regulated by Notch signalling.²⁴ Furthermore, Notch3 receptor, which is predominantly expressed in vascular SMCs, plays a crucial role at every step of intracranial vessels morphogenesis, being involved in the patterning of the CoVW, the arterial differentiation of SMCs and also the postnatal maturation and survival of SMCs.²⁷ Other pathways are also important for vascular morphogenesis, such as Angiopoietin-Tie2 signalling that is essential for vessel maturation and stabilization, and sphingosine-1-phosphate signalling that regulates sprouting angiogenesis through inhibition of VEGF signalling, directs mural cell recruitment and stabilizes endothelial cell-mural cell contacts.^{21,23,28}

A remarkable aspect of these mechanisms is the importance of interactions between the vascular system and cells of the surrounding brain parenchyma in the regulation of brain angiogenesis and the development of the blood brain barrier (see below). This concept is underscored by the 'neurovascular unit'—comprising endothelial cells, pericytes, astrocytes, neurons, and interneurons—which defines an anatomical and functional structure established during early embryogenesis in which each component is intimately and reciprocally linked to each other.¹⁸

3.2 Anatomical and physiological traits peculiar to brain vessels

As is the case in other arteries, SMCs in the cerebral artery wall have two important tasks: (1) mediating (to a large extent) the synthesis and turnover of the collagen, elastin and other ECM proteins, necessary to

maintain the structural integrity of the vessel, and (2) regulating flow in the artery by modulating luminal diameter through vasodilation or vasoconstriction. This enables the intracranial artery to adapt to changes in flow and blood demand, as well as the mechanical stress imposed by blood pressure; it also allows for the repair of damage caused by normal 'wear and tear' or biochemical insults, such as inflammation-related protease activity. Pericytes, the mural cells of capillaries, are generally thought to belong to the same cellular lineage and category as SMCs; however, they exhibit distinct molecular, morphological, and functional features.^{29,30} Notably, pericytes play a key role in blood-brain barrier (BBB) formation and maintenance.³¹

3.2.1 Large arteries as load-bearing conduits

In the artery wall, bundles of collagen fibres are the main load-bearing structures and lamins composed of elastic fibres the main load-distributing component. Unlike other arteries, and especially the extracranial arteries, intracranial arteries lack an external elastic lamina (EEL).^{9,19} Another distinct feature of the large intracranial arteries is that their media layer is discontinuous at major bifurcations, producing so-called medial gaps that lack an elastic lamina and SMCs.^{9,32,33} How much these structural differences affect the elasticity, mechanical strength, and other mechanical properties of the intracranial arteries is not well known. However, it has been shown that the collagen matrix in intracranial artery bifurcations can bear as much load as that elsewhere in the artery, despite the medial gap.³³

Why the structure of the large intracranial arteries differs from that of extracranial arteries is poorly understood, but it may result from differences in vessel development. It is interesting to note that neural crest-derived extracranial arteries do possess an EEL. An essential regulator of the assembly of elastic lamins is signalling via TGF β .³⁴ A TGF β receptor 2 deficiency leads to defective organization of collagen and elastin into the fibres that form bundles and lamins, respectively, with the subsequent end result being vessel dilatation and aneurysm formation.³⁵ Whether differences in the regulation of TGF β signalling in intra- and extracranial vessel development leads to intracranial medial gaps and a lack of EEL remains to be studied.

3.2.2 Small vessels and regulation of cerebral blood flow

Although the brain accounts for only ~2% of body weight, it receives ~20% of the cardiac output, a disproportionate distribution that ensures sufficient oxygenation and delivery of nutrients to neuronal tissue, which is highly metabolically active and very sensitive to hypoxia and hypoglycemia.³⁶ Resting cerebral blood flow (CBF) is primarily determined by perfusion pressure, autoregulatory mechanisms, and the partial pressure of CO₂.³⁷ Arterial pressure fluctuates widely during daily activity. To maintain constant CBF over a substantial range of systemic blood pressures, brain arteries, and arterioles constrict or dilate when intravascular pressure increases or decreases, respectively, a phenomenon referred to as CBF autoregulation.³⁷ CBF is also regulated locally according to the metabolic activity of brain tissue, which fluctuates constantly. The used-dependent increase in CBF to regions where neurons are active (i.e. functional hyperaemia) is mediated by a process termed neurovascular coupling.³⁷ Neurovascular coupling involves complex interactions between neurons, interneurons, astrocytes, and vascular cells within the neurovascular unit, and a number of mechanisms act in parallel to ensure the fidelity of this process. Although multiple lines of evidence support the idea that pericytes play a role in the regulation of CBF,³⁰ a recent elegant study by Grutzlender and colleagues provided strong evidence that

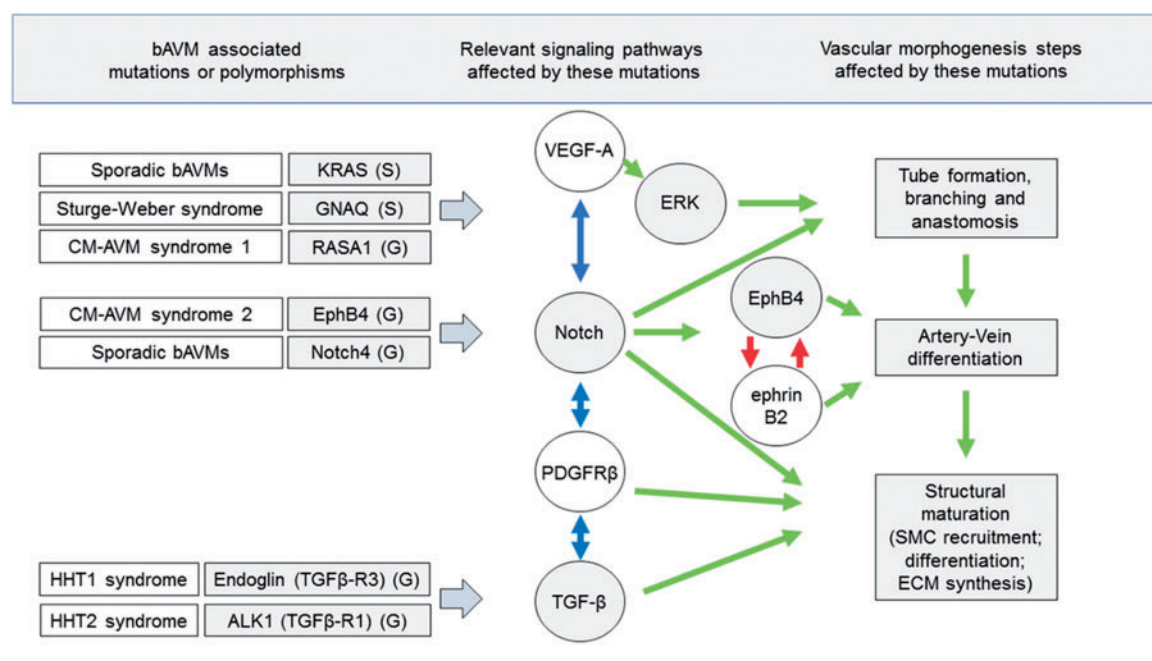


Figure 2 Morphogenesis of a new vessel is an interactive process requiring intercellular communication between endothelial cells, pericytes/SMC, and the local microenvironment. This process is controlled by multiple signalling pathways and growth factors, among which VEGFA, Notch, ephrin B2/EphB4, PDGFB, and TGFβ have a key role. This is demonstrated by the identification of somatic (S) or genomic (G) mutations affecting these signalling pathways (highlighted in grey) in clinical syndromes involving AVM formation. Green arrows indicate induction or upregulation, red arrows inhibition, and blue arrow context dependent interaction. CM-AVM, Capillary Malformation–Arteriovenous Malformation Syndrome; HHT, Hereditary Haemorrhagic Telangiectasia.

neurovascular coupling elicits an active dilatory response exclusively in SMC-covered vascular segments.³⁸ A precise understanding of the mechanisms controlling neurovascular coupling, an area of intense research activity, has been the subject of several recent reviews.^{36,39,40}

The myogenic response, which reflects the intrinsic property of SMCs to constrict in response to increased pressure and relax in response to decreased pressure, is a major contributor to CBF autoregulation.^{37,41} Increases in intraluminal pressure cause a graded depolarization of SMC membrane potential, resulting in activation of voltage-dependent calcium channels and thus increased calcium entry, this activates myosin light chain kinase with the subsequent phosphorylation of the 20-kDa myosin light chain (MLC 20) leading to increased actin/myosin interaction and vasoconstriction.⁴² In addition to calcium entry, signalling mechanisms require calcium-sensitization of the SMC contractile apparatus, which acts primarily through inhibition of myosin light-chain phosphatase, resulting in a calcium-neutral increase in MLC 20 phosphorylation.⁴³ A third important mechanism is actin cytoskeleton reorganization, and the formation of enhanced connections between the actin cytoskeleton, plasma membrane, and ECM to enhance force transmission to the cell membrane.⁴⁴ The RhoA/Rho associated kinase signalling pathway plays a crucial role in these mechanisms via effects on calcium sensitivity and actin dynamics.^{43,44}

The myogenic response sets the level of resting arterial diameter (basal arterial tone) from which other stimuli, including factors released during neuronal activity, can induce vasoconstriction or vasodilation. Hence, the degree of arterial tone determines the changes in diameter elicited by neuronal activation; notably, a diminished myogenic response is associated with decreased functional hyperaemia.^{45,46}

The membrane potential of SMCs has a major influence on the diameter of cerebral arteries. Interestingly, SMCs of penetrating arterioles are

more depolarized at lower pressures than SMCs of pial arteries; accordingly, penetrating arterioles develop more myogenic tone than pial arteries.⁴⁰ Smooth muscle voltage-gated potassium (K_v) channels, inwardly rectifying K^+ channels and large-conductance, calcium-sensitive K^+ channels (which are of lesser importance in penetrating arterioles) control arterial diameter by exerting a major hyperpolarizing influence.⁴⁰ Notably, both endothelial cells and astrocytic endfeet can engage in intercellular signalling to control SMC membrane potential, and thus contractile state, at the arteriolar level.⁴⁰

3.2.3 Capillaries and the blood–brain barrier

Capillaries in the brain are surrounded by pericytes and are encased by endfoot processes of astrocytes, which are interconnected. These connections, together with tight junctions between endothelial cells, form a barrier, termed the BBB that limits the free diffusion of circulating compounds or particles into the neural tissue. Structure and function of the BBB have been extensively reviewed recently and will not be discussed in detail in this review.^{47,48}

4. Involvement of SMCs in diseases that affect intracranial vessels

In the following sections, we summarize key findings highlighting SMC changes in three selected cerebrovascular diseases that highly impact different age groups, namely bAVMs in young adults, IAs in the middle-aged population and SVDs in the elderly, and we discuss the link between their pathogenesis to the characteristic developmental, structural, and functional features of intracranial vessels underlined above.

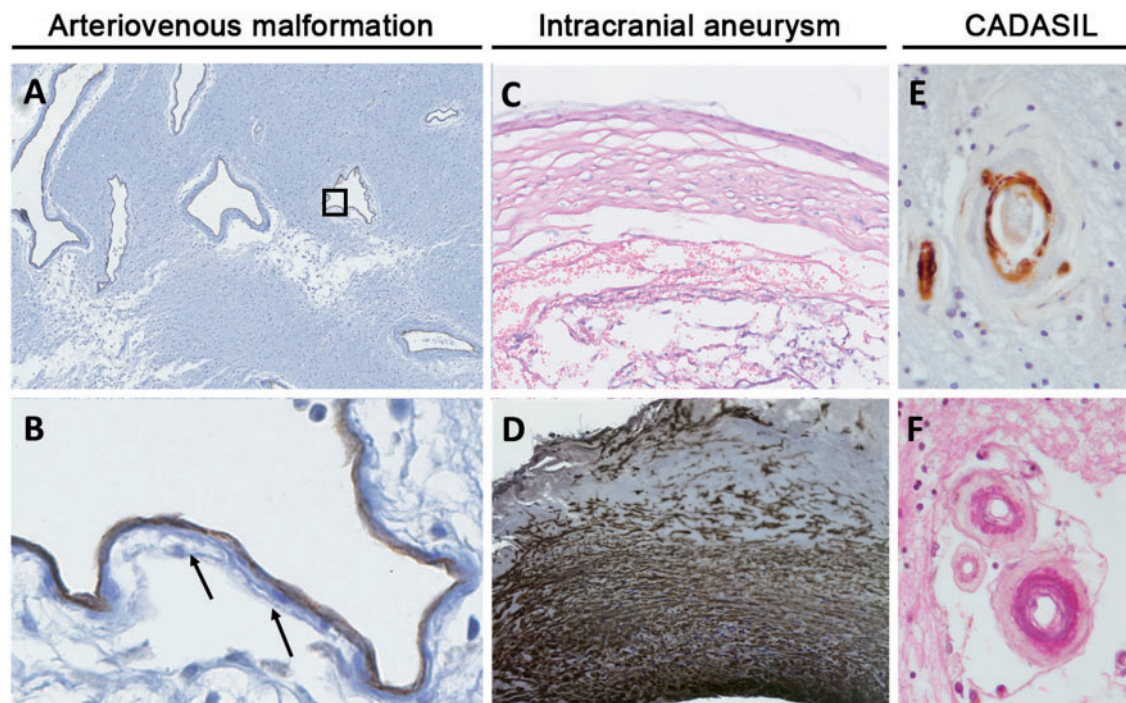


Figure 3 Histopathological features of vessels in bAVM, intracranial aneurysms, and SVD (CADASIL). Representative images of a histological section through a bAVM stained for the endothelial cell marker CD31 (A) showing an increased density of enlarged vessels, between which brain parenchyma is found, characterized by paucity or complete lack of SMCs (arrows in B) (B, higher magnifications of boxed region in A). Representative images of a ruptured IA stained by Hematoxylin and eosin (C) showing major disintegration of the vessel wall with loss of the internal elastica lamina and SMCs, and of an unruptured IA immunostained for α -smooth muscle actin (D) showing in contrast SMC hyperplasia that is also part of the remodelling process. Representative images of brain arterioles from a CADASIL patient immunostained for the SMC marker α -smooth muscle actin (E) or stained by Hematoxylin and eosin (F) showing severe loss of SMCs and thickening of the media by ECM proteins deposition (E and F).

4.1 Arteriovenous malformation—a disorder with altered vascular morphogenesis leading to defective SMC recruitment

bAVMs are a collection of abnormal blood vessels that connect arteries to veins without an intervening capillary bed.⁴⁹

4.1.1 Shunting vessels with few or no SMCs

Aberrant arterio-venous connectivity leads to a high-flow shunt (so-called nidus) that exposes the feeding arteries to abnormally high flow and the draining venous vessels to abnormally high flow and pressure. Shunting vessels in the nidus are also at risk of rupture, since they are not only exposed to abnormally high flow and pressure they also possess a wall structure that resembles that of capillaries⁵⁰ or small arteries, containing few or no SMCs (Figure 3A and B), notwithstanding the amount of hemodynamic stress experienced by these vessels.

4.1.2 Altered vascular morphogenesis

bAVMs may form during prenatal development of the cerebral vasculature, as demonstrated by the occasional diagnosis of these lesions in utero.^{51,52} In line with the developmental origin hypothesis, bAVMs are found in children and young adults, and incidence of bAVM diagnosis does not significantly increase with age.^{12,13} Their nature as an aberration

of brain vasculature formation is further confirmed by transgenic animal models in which mutations linked to bAVM formation in humans predispose to death due to bAVM haemorrhage either after mid-gestation or early in postnatal life.^{53,54} At what stage of development bAVMs start to develop is unknown, and the specific aberration of vascular morphogenesis that leads to bAVMs remains to be defined. Some of the mutations linked with bAVM formation in humans, predispose in transgenic animal models to death due to bAVM haemorrhage either after mid-gestation or early in postnatal life.^{53,54} In addition to the developmental origin, bAVMs can also develop in postnatal life, as demonstrated by case reports of *de novo* bAVM formation in adulthood,⁵² as well as recurrence of bAVMs in long-term follow-up after angiography-confirmed removal.⁵⁵ These observations might be explained by the presence of germline or somatic mutations that affect the control of angiogenesis. In mice, postnatal bAVM development can be induced by constitutive activation of Notch4,⁵⁶ supporting the concept that bAVM formation may occur postnatally due to local dysregulation of angiogenesis.

4.1.3 Imbalance in proangiogenic and antiangiogenic signalling

Given the roles of VEGF, Notch, and TGF β signalling in vascular development (see above), it is not surprising that the identified germline mutations that cause bAVMs in humans affect TGF β signalling,⁵⁷ Notch signalling,⁵⁸ or regulation of RAS-MAPK activity,^{59,60} which is activated in endothelial

cells by VEGFA²² (Figure 2). The somatic mutations reported in GNAQ [Guanine nucleotide-binding protein G(q) subunit alpha] in 80% of Sturge–Weber syndrome associated bAVMs also affect the regulation of RAS-MAPK activity,⁶¹ as do somatic mutations in KRAS, recently discovered in > 50% of sporadic bAVMs.⁶² The activating somatic KRAS mutations found in sporadic bAVMs are enriched in endothelial cells, predisposing to overactive endothelial cell proliferation and migration through increased RAS-ERK1/2 activation.⁶² In addition to mutations activating RAS-MAPK signalling, genetic polymorphisms in Notch4 are associated with bAVMs in humans.⁵⁸ Constitutive expression of Notch4 (or Notch1) in the capillary endothelium of the brain leads to the enlargement of brain capillaries and bAVM formation in mice,⁶³ and Notch4 expression is detected in human bAVMs.⁶⁴ Predisposition to bAVMs caused by mutations in VEGF-A, Notch, or TGFβ signalling in humans, as well as the known roles of these signalling pathways during vascular development suggest that bAVM formation results from an imbalance between angiogenic RAS-MAPK mediated endothelial cell proliferation and TGFβ-mediated recruitment and growth of pericytes or SMCs. The concept that bAVM formation is caused by an imbalance in proangiogenic VEGF signalling and TGFβ signalling, the latter of which promotes mural cell recruitment and vessel maturation, is supported by studies in animal models showing that postnatally induced overexpression of VEGF-A in TGFβ receptor signalling-deficient animals (generated by knockout of the co-receptor ALK1 or TGFβ type I receptor endoglin) leads to formation of bAVMs.^{65,66} From a clinical perspective, the observation that concurrent VEGF antagonism leads to regression of these induced bAVMs is of great interest,⁶⁵ and suggests that anti-VEGF drug therapy might be viable option for the treatment of bAVMs.

4.2 Intracranial aneurysms—a flow-driven remodelling of the large load-bearing conduit arteries

IAs are not innate lesions, but instead form at bifurcations of the large intracranial arteries near the CoW during postnatal life, usually after the age of 40 years (Figure 1).^{8,9}

4.2.1 Arterial remodelling of the large load-bearing conduit arteries with SMCs loss

The loss of SMCs is characteristic of the arterial remodelling that initiates formation of IA as well as its rupture (Figures 3C and 4).^{9,67–69} The crucial role of SMCs in maintaining the structural integrity of the cerebral artery wall is further supported by the reported case of a child carrying a mutation in myosin heavy chain 11 (MYH11) who developed rapidly progressing IAs at a very early age, one of which ruptured.⁷⁰ Not all IAs that form ultimately rupture⁹ (Figure 4). IA rupture occurs when the ECM of the IA wall is too weak to withstand the mechanical load imposed by blood pressure. This may occur when the IA wall has lost its SMCs and, consequently, the capacity to repair and adapt (Figures 3C and 4).^{9,67} Balance between the adaptive wall remodelling and hemodynamic stress also determines whether the IA will remain stable or grows (Figure 4), since IA growth is not possible without remodelling of the collagen that does not stretch.⁷¹ Available evidence suggests that the main cause for this loss of SMCs from the IA wall is the accumulation of lipids in the IA wall followed by lipid uptake by SMCs, a process that turns SMCs into foam cells.^{72,73} Subsequently the physiological function of these SMCs turned foam cells is impaired, and the eventual death of many of these foam cells reduces the number of cells in the IA wall, and consequently the capacity to synthesize new ECM.^{9,67,72,73} This lipid accumulation

occurs irrespective of serum lipid levels in all IA walls to some extent, probably because of damaged or dysfunctional endothelium, which in turn is damaged by non-physiological flow in the IA lumen.^{9,67}

4.2.2 Flow-driven remodelling of the cerebral artery wall

Environmental or acquired risk factors are considered the main drivers of the disease.⁷⁴ The following observations support the concept that flow-driven remodelling of the cerebral artery wall plays a critical role in IA formation: (1) IA formation can be induced in laboratory animals by procedures that manipulate the cerebral circulation and increase flow in intracranial arteries;^{75,76} (2) IA formation is observed in the feeding arteries of bAVMs—arteries that have abnormally high flow owing to the direct arteriovenous shunt;^{77,78} (3) IAs that form in feeding arteries of bAVMs may spontaneously thrombose and disappear as the bAVM is treated;⁷⁸ and (4) IA formation typically occurs at bifurcations exposed to flow conditions that produce high wall shear stress.⁷⁹ Flow conditions, and especially the magnitude and distribution of flow-induced mechanical stress, including wall shear stress, are determined to a large extent by vessel geometry.^{80,81}

4.2.3 Developmental variations of the CoW, a predisposing factor

Although IAs are not formed during the formation and development of the intracranial vasculature, variations in cerebrovascular development can predispose to later development of the lesions. In fact, in only ~52% of cases does the CoW have a 'normal' anatomy, with the most common anatomical variation being hypoplasia of one or more of the arteries that form the CoW (Figure 1).^{82,83} Such variations remain asymptomatic unless the adequate function of the intracranial vasculature is challenged by other pathologies, but can also lead to flow conditions that trigger arterial remodelling and eventually lead to IA formation. Historically, fenestrations of intracranial arteries and artery duplications have indeed been associated with the presence of IAs, although data on the topic is limited.⁸³ Of considerable interest in this context is the more recent work by Bourcier et al.,⁸⁴ who compared the geometry of the major arteries in the anterior communicating artery complex between familial IA patients, their non-affected first degree relatives, and patients with sporadic IAs. The anterior communicating region is one of the most common locations for ruptured sIAs, and for the formation of IAs overall,⁸⁵ possibly because of the complex flow conditions produced by the H-configuration of the arterial anastomosis⁸⁶ and the very frequent asymmetry of the involved anterior cerebral arteries. The findings by Bourcier et al.⁸⁴ suggested that the angle between A1 and A2 segments, as well as the asymmetry of the anterior cerebral artery, not only predispose to the formation of IAs overall but may explain IA formation in some patients with familial predisposition to the disease.

4.2.4 Mechanistic clues coming from genetics

IA usually occurs as a sporadic condition, although a history of aSAH or IA among relatives is clearly a risk factor for IA formation, suggesting that genetic factors are involved in the development of IAs.⁹ It seems, however, that the contribution of genetic polymorphisms to the risk of IA development in the general population is relatively low, although it may affect the development and eventual outcome of sIAs.⁷⁴ The observation that the concordance of IA locations is higher in monozygotic than dizygotic twins suggests a genetic influence also on the location of IA formation.⁸⁷ Furthermore, the risk attributable to IA associated genetic polymorphisms is different in the middle cerebral artery bifurcation than elsewhere in the cerebral vasculature (most other IAs reside in the

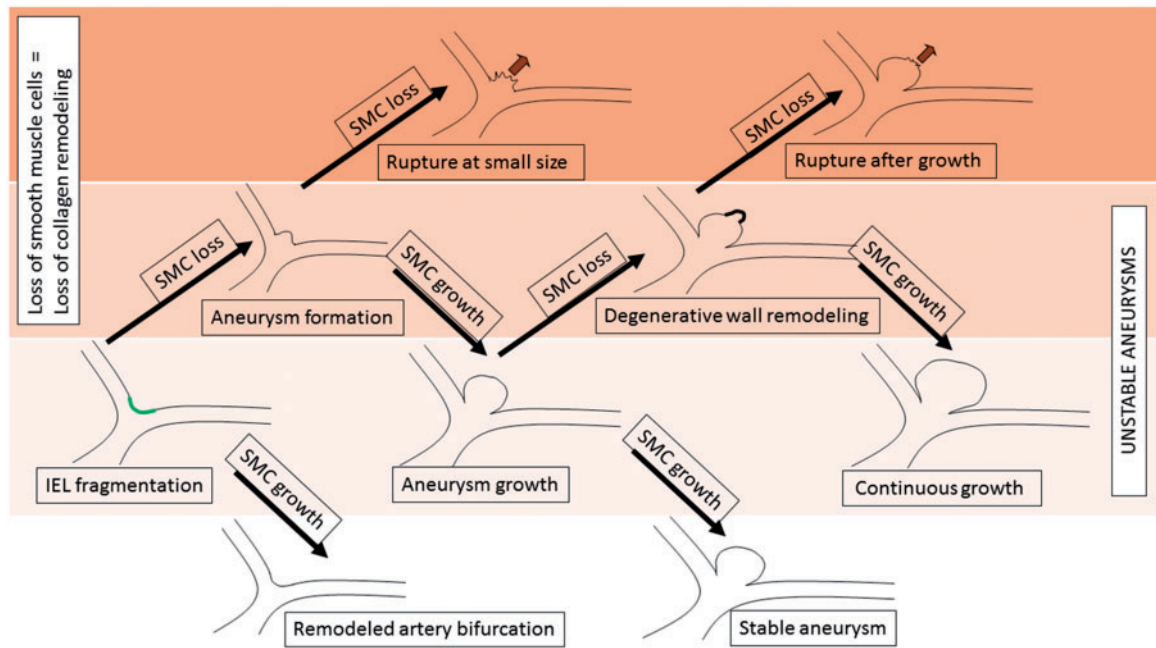


Figure 4 Role of SMCs in the clinical course of IAs. Loss of SMCs, manifesting as a significant reduction in their number or as their complete loss, is characteristic of the formation and progression of IAs (as shown in Figure 3C). After formation some IAs, however, develop a thickened wall characterized by SMC hyperplasia (see Figure 3D), which probably reflects an adaptive remodelling response that responds to the wall tension imposed by hemodynamic stress. Some IAs start to grow, not because of passive distension because collagen does not distend,⁷¹ but because of active remodelling of the collagen matrix by mural fibroblasts, myofibroblasts, and SMCs proliferation. As a result of IA growth, the flow conditions in the IA lumen change, which in turn affects the wall biology through the flow sensitive endothelium. This may trigger a degenerative remodelling process leading to loss of SMC through lipid accumulation.

CoW; see Figure 1).⁸⁸ Considering the different ontogeny of neural crest-derived SMCs in the anterior circulation vs. mesoderm-derived SMCs in the posterior circulation,¹⁵ as well as the genetic regulation of the morphogenesis of the CoW,²⁷ these findings could imply that some IA- and aSAH-associated genetic polymorphisms affect the morphogenesis of intracranial artery bifurcations or exert effects that depend on IA location in the cerebrovascular tree.

There has been an intense search for genetic polymorphisms that predispose to IA formation or aSAH.⁸⁹ Of the numerous loci and alleles reported to be associated with IA disease, the strongest associations replicated in different cohorts and populations are found for (1) the *CDKN2B* (cyclin dependent kinase inhibitor 2B) gene, (2) the transcription regulator gene, *SOX17* (SRY-box 17); and (3) the *EDNRA* (endothelin receptor A) gene.⁸⁹ Of these, *EDNRA* encodes a protein known to mediate the genesis of pharyngeal arteries from the neural crest—one of the initial steps in the formation of the ICA and cerebral vessels (see above).^{90,91} The *EDNRA* allele rs68415181 associated with IAs in humans represses transcriptional activity of *EDNRA*,⁹² and SMC-specific inactivation of *Ednra* in mice induces abnormal vascular development.⁹³

4.3 Small-vessel diseases—a degenerative process affecting small intracranial vessels that compromises CBF regulation and BBB

4.3.1 Amyloid and non-amyloid SVDs

Cerebral amyloid angiopathy (CAA) and non-amyloid SVD are the most prevalent SVDs that can manifest as sporadic or hereditary conditions.⁴

Cerebral amyloid angiopathy (CAA) typically affects small arteries and arterioles (CAA type 2) or capillaries (CAA type 1) in the cerebral cortex and overlying meninges, and is commonly associated with multiple lobar cerebral microbleeds, spontaneous intracerebral haemorrhages, as well as cortical superficial siderosis. Beside haemorrhagic lesions, CAA is also associated with ischaemic lesions including cortical microinfarcts and white matter lesions.⁹⁴ In contrast, non-amyloid SVD predominantly affects the small penetrating vessels (arterioles, capillaries, and sometime venules) within the white matter and the deep parts of the brain, including basal ganglia, thalamus, and brainstem, and is usually associated with lacunar infarcts, macro or microhaemorrhages in these deep brain regions, as well as diffuse white matter lesions.⁴

Autopsy studies indicate that advancing age is the strongest risk factor for sporadic CAA, with any form of CAA occurring in ~25% of adults over the age of 70 years and 55% of adults over the age of 90 years.^{95,96} Age is also a strong risk factor for non-amyloid SVD, with moderate-to-severe white matter lesions being detected on brain MRI in 26% of adults over the age of 65 years in a pooled population-based study.⁹⁷ Traditional vascular risk factors including hypertension, smoking, and diabetes are commonly associated with features of non-amyloid SVD,⁴ although the association between vascular risk factors and SVD is not completely understood.⁹⁸ Moreover, a recent study suggests that they actually explain only a small proportion of the variance in SVD features.⁹⁹

4.3.2 Degeneration of SMC

CAA is characterized by progressive accumulation of amyloid β ($A\beta$) peptide, mostly the $A\beta$ 40 species, within the vessel wall along mural cell

basement membranes, and simultaneous loss of SMCs; secondary vessel changes may include fibrinoid necrosis and microaneurysm.¹⁰⁰ CAA is mainly driven by reduced A β peptide clearance. Disruption of peri- and paravascular drainage pathways, possibly as a result of reduced arterial vasomotion and changes in vascular basement membrane composition are increasingly recognized as important factors in this process.¹⁰¹ Although it has been frequently proposed that the deposition of A β in the vessel wall makes these vessels fragile and causes their rupture, a recent study showed that sites of haemorrhages were associated with reduced rather than increased vascular amyloid burden.¹⁰² In non-amyloid SVD, vessel changes are also characterized by progressive degeneration and loss of SMCs sometimes with infiltration of plasma components or lipids in the vessel wall and fibrinoid necrosis.¹⁰³ Interestingly, CADASIL (Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy)—the most frequent hereditary SVD—is caused by dominant mutations in the extracellular domain of the Notch3 receptor, which is predominantly expressed in vascular SMCs and brain pericytes, thereby establishing that SVD features can be primarily driven by mural cell alterations (Figure 3E and F).¹⁰⁴ Yet, the exact mechanisms of SMC degeneration in non-amyloid SVD as well the mechanism by which SMC loss ultimately causes intracerebral haemorrhages or ischaemic lesions in cerebral SVDs (see below) are not completely understood. An intriguing observation is the scarcity of intracerebral haemorrhage in CADASIL or CARASIL (Cerebral Autosomal Recessive Arteriopathy with Subcortical Infarcts and Leukoencephalopathy), two hereditary SVDs with major arterial SMC loss.¹⁰³

4.3.3 BBB dysfunction

BBB leakage has been especially well documented in patients and mice with CAA. Deposition of fibrinogen and serum proteins around affected capillaries has been shown in post-mortem tissues from patients with CAA.¹⁰⁰ Moreover, leakage of blood derived molecules and intravenously injected exogenous tracers has been shown in several mouse models overexpressing A β .¹⁰⁵ A β is highly toxic to capillary endothelial cells and pericytes, which are important for BBB integrity, and pathological studies have shown aberrant expression of endothelial transporters or receptors, loss of tight junctions as well as degeneration of endothelial cells and pericytes. These changes can in turn further diminish A β clearance, accelerate A β deposition and aggravate CAA.¹⁰⁵ The importance of BBB alterations in the development of brain lesions is highlighted by the analysis of brain pericyte deficient mice that exhibit reduction in brain microcirculation, chronic hypoperfusion, and secondary neuronal injury and neuroinflammation.¹⁰⁶

4.3.4 Vessel wall remodelling

Another key feature of SVD, especially in non-amyloid SVD, is the major remodelling of the ECM leading to prominent thickening of vessel wall.¹⁰³ These changes have been predominantly post-mortem observations in SVD patients (i.e. at the end stage of the disease), and some of these changes have thus far been interpreted as secondary events in response to SMC degeneration that serve as a repair mechanism.¹⁰⁷ There is ample evidence that sustained elevations of blood pressure at an early stage alter the structure of small brain vessels, leading to a smaller lumen, vascular stiffening, and increased vascular resistance as a result of vessel wall reorganization (eutrophic inward remodelling) or SMC hypertrophy (hypertrophic remodelling).¹⁰⁸ These defects are of particular

importance because they can adversely affect local CBF by decreasing microvascular pressure and maximal vasodilatory capacity. Inward remodelling of cerebral arteries is also an early disease feature in CADASIL mice. Notably, this process occurs without elevated blood pressure as a consequence of increased Notch3 activity in SMCs, suggesting the involvement of other factors than hypertension.¹⁰⁹

4.3.5 Neurovascular dysfunction

It is generally assumed that infarcts and white matter lesions are of ischaemic origin, reflecting vessel wall remodelling or structural alterations of the vasomotor apparatus, involving damage to SMCs, and transformation of penetrating arteries into the so-called 'earthen pipe' state, in which neither dilation nor constriction is possible.¹¹⁰ Although this is very likely true at the final stage of the disease, multiple lines of evidence suggest that functional alterations of cerebral vessels and more subtle structural vessel changes that occur early disrupt key cerebrovascular mechanisms that may underlie brain lesions. With respect to CAA, experimental studies using either infusion or topical application of A β or transgenic mice overproducing A β , have shown that A β impairs functional hyperaemia, CBF autoregulation, and endothelium-dependent vasodilation. These alterations are mediated by the CD (cluster of differentiation) 36 receptor, to which A β binds, leading to NADPH-dependent production of reactive oxygen species.^{111–113} Recent work from the Iadecola's lab revealed that the effector cells exposed to A β and responsible for the production of reactive oxygen species in the neurovascular unit are the brain perivascular macrophages, a distinct population of resident brain macrophages closely associated with the abluminal wall of arterial SMCs.¹¹⁴ Oxidative-nitrosative stress causes DNA damage in endothelial cells, which in turn activates transient receptor potential melastin-2 channels leading to intracellular calcium overload and endothelial dysfunction.¹¹⁵ With respect to sporadic non-amyloid SVD, experimental studies have established that chronic hypertension can similarly alter functional hyperaemia, CBF autoregulation, and endothelium-dependent vasodilatory responses. Likewise, NADPH-derived reactive oxygen species have emerged as key mediators of hypertension-related neurovascular dysfunction and brain perivascular macrophages have also been implicated as key effector cells in this effect.^{116,117} In CADASIL, a genetic paradigm of non-amyloid SVD, experimental studies have highlighted SMCs as key effector cells of neurovascular dysfunction. Impaired functional hyperaemia and CBF autoregulation as well as attenuated responses to both endothelium-dependent and SMC-vasodilators are early and prominent features in transgenic mice expressing an archetypal CADASIL mutation, despite the absence of overt SMC loss.¹¹⁸ Studies by us and our collaborators have established that diminished myogenic responses of cerebral arteries contribute to cerebrovascular dysfunction in CADASIL and are caused by upregulation of voltage-gated potassium (K_V) channels at the plasma membrane of SMCs.¹¹⁹ We further identified a novel mechanism underlying this channelopathy-like defect, showing that TIMP3 (tissue inhibitor of metalloproteinases-3), a protein tightly bound to the ECM of brain arteries, forms complexes with extracellularly deposited Notch3 extracellular domains and abnormally accumulates in the media of brain arteries of CADASIL patients and CADASIL model mice. The resulting increase in TIMP3 activity inhibits the metalloproteinase, ADAM17, decreasing ADAM17-dependent shedding of the epidermal growth factor receptor (EGFR) ligand, HB-EGF, thereby diminishing EGFR-mediated endocytosis of K_V channels. Thus, increased levels of TIMP3

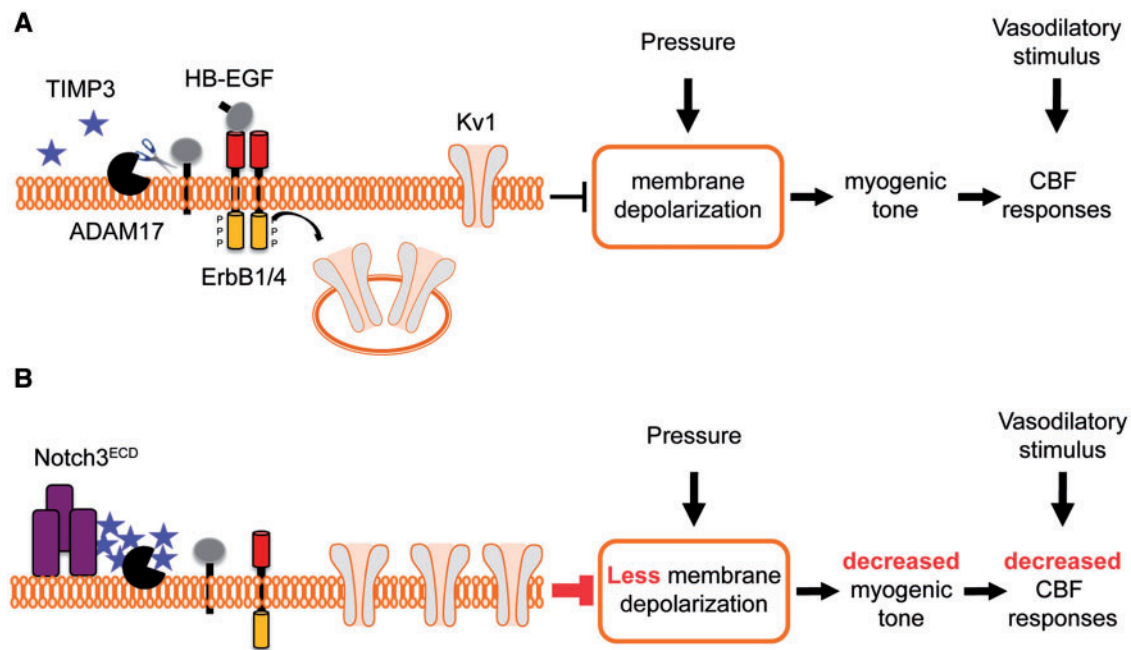


Figure 5 Presumed signalling pathways through which NOTCH3 mutations impair cerebrovascular function in CADASIL. (A) Physiological conditions (upper panel), TIMP3 is present in a low abundance in the ECM of brain arteries and ADAM17 at the cell surface of SMCs is therefore active, resulting in the release of the soluble form of HB-EGF with subsequent ErbB1/ErbB4 activation and K_v1 channel endocytosis. This relieves the tonic hyperpolarizing influence of these channels on the membrane potential of SMCs, thereby allowing full development of myogenic tone of brain arteries and enabling full evoked CBF responses. (B) In CADASIL (lower panel), Notch3^{ECD} accumulates at the surface of SMCs, leading to an increase in the amount of TIMP3, which binds to and inhibits ADAM17 resulting in increased K_v1 current density that acts as a brake to limit the development of myogenic tone and evoked CBF responses. From ref. ⁴⁶, licensed under a Creative Commons Attribution license.

blunts the ADAM17—HB-EGF—EGFR— K_v pathway in cerebral arterial myocytes to attenuate the myogenic tone of cerebral arteries and compromise CBF regulation in CADASIL (Figure 5).^{46,118} Interestingly, myogenic responses and density of K_v channels are unaltered in peripheral arteries (mesenteric arteries), and K_v channels do not seem to be regulated by HB-EGF—EGFR in these arteries, suggesting a possible mechanistic basis for the exclusive cerebrovascular manifestations in CADASIL.¹¹⁹

4.3.6 ECM changes, a proximate cause in SVDs

The novel ECM-dependent mechanism identified in CADASIL suggests that perturbations of the microvascular ECM are a proximate cause in SVDs.¹²⁰ In support of this novel concept, most monogenic forms of SVDs are caused by mutations in genes encoding ECM or ECM-associated proteins. These include dominant mutations in the coding region of $\alpha 1$ (COL4A1) or $\alpha 2$ (COL4A2) chains of collagen type IV, a major component of vascular basement membranes, in COL4A1/COL4A2-related haemorrhagic stroke; dominant gain-of-function mutations in the 3' untranslated region of COL4A1 in PADMAL (Pontine Autosomal Dominant Microangiopathy with Leukoencephalopathy); recessive and dominant loss-of-function mutations in HTRA1, a serine protease that regulates the TGF β pathway, in CARASIL and HTRA1-related SVDs, respectively; and dominant mutations in cathepsin A in CARASIL (Cathepsin A-Related Arteriopathy with Strokes And Leukoencephalopathy).^{121–124}

5. Summary

Cerebrovascular diseases are among the most significant causes of morbidity and major drivers of healthcare costs in the developed world. Although symptoms caused by cerebrovascular diseases are attributable to damage to neurons or glial cells, the underlying problem often lies in the vessels and is primarily found in SMCs or in cell–cell interactions involving SMCs.

In bAVMs, dysregulated morphogenesis of vessels leads to large, high flow vessels that lack adequate SMC layer and remain capillary-like, predisposing them to rupture under high hemodynamic stress. In IAs, loss of SMCs is part of the remodelling process leading to lesion formation and rupture as well as the degenerative remodelling that may render a previously stable IA unstable and rupture-prone again. Loss of SMCs or alteration of their function is also at the heart of SVDs.

The embryonic origin, structure, and function of intracranial vessels are distinct from those of extracranial vessels, suggesting that SMCs of intracranial vessels have unique physiological properties. Hence, intracranial vasculature warrants greater research attention in the future (1) to better understand the molecular mechanisms involved in its morphogenesis, (2) to further our understanding of the interaction between flow and vessel wall, (3) to identify disease-associated somatic mutations that may occur in utero or postnatally and may affect only the intracranial vasculature, or specific parts of it, (4) to determine how acquired stress factors such as lipid accumulation and inflammation affect the physiology and function of SMCs in different parts of the intracranial

vasculature, (5) to elucidate the mechanisms of SMC degeneration at work in non-amyloid SVD, and (6) to characterize microvascular ECM with respect to vessel type and location, in both physiological and pathological settings and assess the contribution of ECM changes in SVDs. This will expand the basic scientific knowledge necessary to develop solutions to one of the most significant healthcare problems affecting the developed world in our time.

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